Assignment: 2D Crystallographic Point Groups

Condensed Matter Physics

The basic idea

The problem is to find all the point groups compatible with a 2D periodic lattice. The key constraint comes from the fact that any rotation $R(\theta)$ must map lattice vectors to lattice vectors. In the lattice basis, R has integer entries, so its trace must be an integer [1, 2]:

$$\operatorname{tr} R = 2\cos\theta \in \mathbb{Z}$$

This only works for $\cos\theta \in \{0, \pm\frac{1}{2}, \pm 1\}$, which gives $\theta = 0°, 60°, 90°, 120°, 180°$, or equivalently $n = 1, 2, 3, 4, 6$. So pentagons are out – you simply can't tile a plane with 5-fold symmetry and keep a lattice [1].

The full list in Schoenflies notation is [3, 4]:

$$C_1,\ C_2,\ C_3,\ C_4,\ C_6 \quad \text{(rotation only)}$$

$$D_1,\ D_2,\ D_3,\ D_4,\ D_6 \quad \text{(rotation + mirrors)}$$

That's 10 point groups total.

How I'm constructing them: the motif trick

In class we did the dice thing – you place a motif at every Bravais lattice point, and the symmetry of the whole pattern is determined by *both* the lattice *and* how you shade (mark) the motif [5]. The idea is:

- Put dots **off** all mirror lines → no mirrors survive → you get a cyclic group C_n.
- Put dots **on** the mirror lines symmetrically → mirrors preserved → you get a dihedral group D_n.

Each Bravais lattice has a maximum point group symmetry, and by shading you can realise any subgroup of that [2, 5]:

Lattice	Max symmetry	Motif
Oblique	C_2	parallelogram
Rectangular	D_2	rectangle
Centred rectangular	D_2	rectangle
Square	D_4	square
Hexagonal	D_6	hexagon / triangle

Below is a summary figure (Fig. 1) of all 10 groups, then I go through each one.

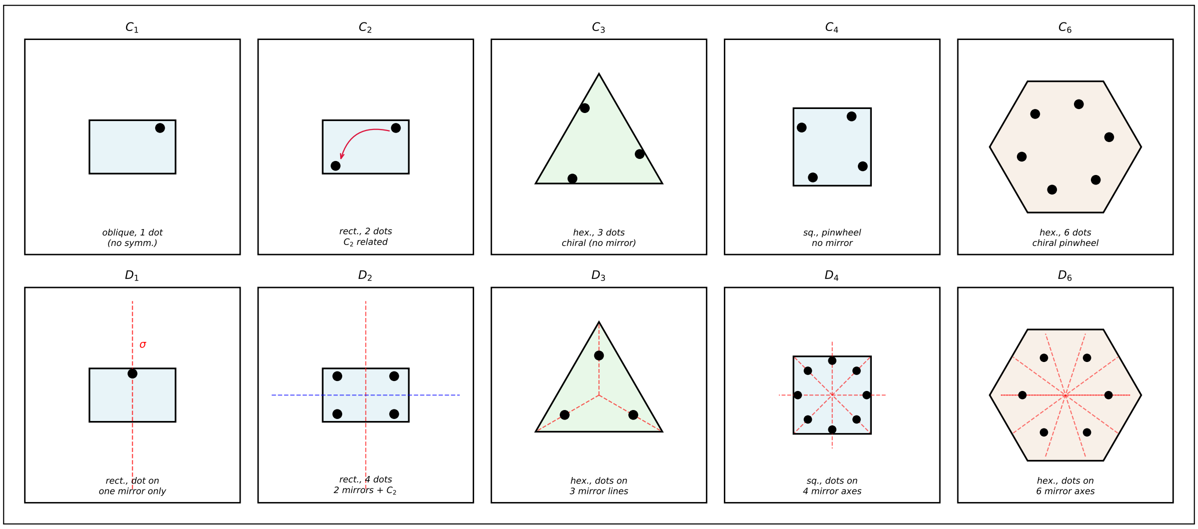

Figure 1: All 10 two-dimensional crystallographic point groups illustrated using the lattice+motif (shaded-basis) method [1, 5]. Each panel shows a motif (rectangle / square / triangle / hexagon) placed at a Bravais lattice point; filled circles indicate the shading of the motif. *Top row* (C_1–C_6): cyclic groups – dots placed *off* all mirror lines, so only n-fold rotation survives. *Bottom row* (D_1–D_6): dihedral groups – dots placed *on* the mirror axes (red dashed lines), preserving both rotations and reflections. Figure generated with Python/Matplotlib [7]; script: `generate_fig1.py` (provided separately).

Going through each group

C_1 – no symmetry (identity only)

Take an oblique lattice and put a single dot at a completely random position inside the motif – not on any axis, not on any mirror. The only thing that maps this pattern to itself is the identity (360° rotation). No C_2, no mirrors. This is the "trivial" point group, everything has at least C_1 symmetry [4].

$$\boxed{\text{Oblique lattice} + \text{single asymmetric dot} \;\Rightarrow\; C_1}$$

C_2 – 2-fold rotation, no mirrors

Use a rectangular lattice. Place two dots inside the rectangle, related to each other by a 180° rotation about the centre. The crucial thing is to put them *off* the horizontal and vertical mirror lines – if either dot sat on a mirror, you'd have D_1 or D_2 instead. With this asymmetric placement the C_2 rotation maps dot 1 to dot 2 and back, but no reflection does [1]. Group order = 2.

$$\boxed{\text{Rectangular lattice} + \text{two dots related by } C_2\text{, off all mirrors} \;\Rightarrow\; C_2}$$

C_3 – 3-fold rotation, no mirrors

Switch to a hexagonal (triangular) lattice. Use a triangular motif and place three dots arranged in a small *chiral* triangle inside it – i.e. the inner triangle is rotated relative to the outer one, so none of the three dots sits on any altitude (which would be a mirror line). Each dot maps to the next by 120°. No mirror survives [5].

$$\boxed{\text{Hex. lattice} + \text{chiral 3-dot motif} \;\Rightarrow\; C_3}$$

C_4 – 4-fold rotation, no mirrors

Square lattice, square motif. Arrange four dots in a pinwheel pattern – like a swastika shape – each related to the next by 90°. The key: rotate the whole pattern by 45° so that none of the dots lands on the four mirror lines of the square (x-axis, y-axis, and the two diagonals). The C_4 axis is there, all four mirrors are broken [2]. Group order = 4.

$$\boxed{\text{Square lattice} + \text{pinwheel 4-dot motif} \;\Rightarrow\; C_4}$$

C_6 – 6-fold rotation, no mirrors

Hexagonal lattice, hexagonal motif. Six dots in a pinwheel: each is related to the next by 60°, but the whole arrangement is rotated slightly so none of the dots sits on any of the six mirror lines of the hexagon [5]. Group order = 6.

$$\boxed{\text{Hex. lattice} + \text{pinwheel 6-dot motif} \;\Rightarrow\; C_6}$$

D_1 – one mirror, no rotation (other than identity)

Rectangular lattice again. Now place a single dot exactly on the *vertical* mirror line of the rectangle, but *not* on the horizontal one. The vertical mirror maps the dot to itself (fine). The horizontal mirror would move the dot to a different height – broken. The C_2 would put a second dot at the mirror image through the centre – also broken. So we're left with just one mirror. This is the same as C_{1v} in some notations [3]. Group order = 2.

$$\boxed{\text{Rect. lattice} + \text{dot on one mirror only} \;\Rightarrow\; D_1}$$

D_2 – 2-fold rotation + 2 mirror lines

Rectangular lattice. Place four dots, one in each quadrant, symmetrically: (x_0, y_0), $(-x_0, y_0)$, $(x_0, -y_0)$, $(-x_0, -y_0)$. Now both mirrors (horizontal and vertical) are preserved, and the C_2 rotation is also a symmetry (it maps each dot to the one diagonally opposite). The full symmetry of the rectangle is restored [1]. Group order = 4.

$$\boxed{\text{Rect. lattice} + \text{4 dots symmetric in both mirrors} \;\Rightarrow\; D_2}$$

D_3 – 3-fold rotation + 3 mirror lines

Hexagonal lattice, triangular motif. Place one dot on each of the three altitude lines of the equilateral triangle (the mirror lines). The C_3 rotation cycles the three dots, and each of the three mirrors maps the dot on its line to itself and swaps the other two. Everything is preserved. This group is isomorphic to S_3, the symmetric group on 3 elements [3, 4]. Group order = 6.

$$\boxed{\text{Hex. lattice} + \text{3 dots on the 3 altitudes} \;\Rightarrow\; D_3}$$

D_4 – 4-fold rotation + 4 mirror lines

Square lattice. The square has 4 mirror lines: the x-axis, y-axis, and the two diagonals. Place dots symmetrically on all four of them (one dot per mirror axis, all at the same radius from centre). The C_4 rotation cycles the dots on adjacent mirrors, so it's a symmetry. All 4 mirrors are preserved [2]. Group order = 8.

$$\boxed{\text{Square lattice} + \text{dots on all 4 mirror axes} \;\Rightarrow\; D_4}$$

D_6 – 6-fold rotation + 6 mirror lines

Hexagonal lattice, hexagonal motif. The hexagon has 6 mirror lines (3 through opposite vertices, 3 through midpoints of opposite edges). Place one dot on each mirror line at equal radius. The C_6 rotation cycles all 6 dots, and all 6 mirrors are preserved. This is the largest 2D crystallographic point group [5]. Group order = 12.

$$\boxed{\text{Hex. lattice} + \text{dots on all 6 mirror axes} \;\Rightarrow\; D_6}$$

Summary

Group	n-fold rotation	No. of mirrors	Order	Lattice
C_1	1	0	1	Oblique
C_2	2	0	2	Oblique / rectangular
C_3	3	0	3	Hexagonal
C_4	4	0	4	Square
C_6	6	0	6	Hexagonal
D_1	1	1	2	Rectangular
D_2	2	2	4	Rectangular
D_3	3	3	6	Hexagonal
D_4	4	4	8	Square
D_6	6	6	12	Hexagonal

A few things worth noting:

- $C_n \subset D_n$ always – the dihedral group contains the cyclic group as a subgroup (the rotations).
- The order of D_n is $2n$ because you have n rotations and n mirror reflections.
- $C_5, C_7, C_8, \ldots$ are perfectly fine abstract groups, but they are *not* crystallographic – no periodic lattice can support them, because $2\cos(2\pi/5)$ and similar values are irrational, not integers [1, 2].
- Quasicrystals (e.g. the Al-Mn alloy discovered by Shechtman in 1982 [6]) *do* show 5-fold diffraction patterns, but they are not periodic lattices – they are quasiperiodic.

References

[1] C. Kittel, *Introduction to Solid State Physics*, 8th ed. Wiley, Hoboken, NJ, 2005. (Ch. 1: Crystal Structure; crystallographic restriction and Bravais lattices.)

[2] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*. Holt, Rinehart and Winston, New York, 1976. (Ch. 7: Classification of Bravais lattices and crystal structures.)

[3] C. J. Bradley and A. P. Cracknell, *The Mathematical Theory of Symmetry in Solids*. Clarendon Press, Oxford, 1972. (Complete tables of point groups in Schoenflies and international notation.)

[4] G. Burns and A. M. Glazer, *Space Groups for Solid State Scientists*, 2nd ed. Academic Press, San Diego, 1990. (Ch. 2: Point groups and their relation to lattice symmetry.)

[5] C. Hammond, *The Basics of Crystallography and Diffraction*, 4th ed. Oxford University Press, Oxford, 2015. (Ch. 2–4: The lattice+motif construction for point and space groups.)

[6] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, "Metallic phase with long-range

orientational order and no translational symmetry," *Phys. Rev. Lett.*, **53**, 1951 (1984). (Discovery of icosahedral 5-fold symmetry in Al–Mn quasicrystals.)

[7] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, **9**(3), 90–95 (2007). (Fig. 1 generated using Matplotlib; script `generate_fig1.py`.)